

Comprehensive Geophysical Characteristics and Exploration Significance of the Xiarihamu Super-Large Copper-Nickel Deposit in the East Kunlun Metallogenic Belt

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Abstract: This study systematically summarizes the key geophysical exploration techniques and their application effectiveness in the exploration of the Xiarihamu super-large copper-nickel sulfide deposit in the Eastern Kunlun metallogenic belt. During the exploration of this deposit, geophysical methods have always played an indispensable role. Recent exploration campaigns have further identified multiple mineralized intrusions surrounding the main orebody. To optimize geophysical methodologies for deep and peripheral exploration, thereby supporting the new round of strategic mineral exploration initiatives and enhancing reserve expansion and production capacity, this research focuses on the geophysical characteristics of the Xiarihamu large copper-nickel sulfide deposit. Comprehensively reviewing the integrated geophysical exploration conducted within the region, this study systematically summarizes the characteristics of geophysical responses. Specifically, it analyzes the results from regional high-precision ground magnetic surveys and various geophysical profiling methods implemented along the same exploration line L7 as the main orebody, including high-precision magnetic, induced polarization (IP), high-precision gravity, Controlled Source Audio-Frequency Magnetotellurics (CSAMT), and Transient Electromagnetic (TEM) methods. The investigation evaluates the effectiveness and applicability of these geophysical approaches for exploring copper-nickel sulfide deposits. The research provides fundamental geophysical methodology references and practical recommendations for both this specific deposit and similar types of mineralization, while clarifying the exploration significance of different geophysical techniques in such deposit exploration.

Keywords: Eastern Kunlun metallogenic belt; Xiarihamu Super-Large Copper-Nickel Deposit; Comprehensive Geophysical Characteristics; Deep and Peripheral Exploration

Introduction

The Xiarihamu Copper-Nickel Deposit was newly

discovered and yielded significant breakthroughs during the implementation of the "358 Geological Exploration Project" in Qinghai Province. The discovery of this deposit marks the first identification of a magmatic

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sulfide deposit with million-ton grade reserves worldwide since the 1994 discovery of the Voisey's Bay deposit in the Labrador region of Canada (Naldrett et al., 2000), and geophysics has played a major role in the discovery, delineation, and understanding of Voisey's Bay (Crebs, 1996). It is the second-largest giant copper-nickel (Cu-Ni) sulfide deposit in China, second only to the Jinchuan deposit (Li et al., 2023a; Porter, 2016). Furthermore, the Xiarihamu deposit represents the first discovery of a world-class super-large magmatic segregation-type Cu-Ni sulfide deposit within the East Kunlun Metallogenic Belt (Pan, 2015). This breakthrough in terms of metallogenic type, mineral species, and deposit scale has filled a critical gap in Cu-Ni ore exploration along the East Kunlun Metallogenic Belt in Qinghai Province, while also ending the historical absence of large-scale nickel deposits in the region. It serves as both a significant achievement and a representative example of the theoretical framework and practical application of the 'Small Intrusions, Giant Deposits (SIGD)' model (Li et al., 2012; Tang et al., 2015).

Recent exploration efforts have delineated 26 intrusive bodies in the peripheral zones of the main orebody, with 10 of these exhibiting mineralization, demonstrating considerable exploration potential (King and Milkereit, 2007; Ma et al., 2024; Wang et al., 2024; Yan et al., 2016). However, the deposit is situated in a high-altitude desert plateau region above 4,000 meters, characterized by extensive Quaternary overburden and limited surface mineralization indicators, posing significant challenges to comprehensive geophysical exploration across the entire area. Especially, the exploration at this mining area has entered a stage focused on deep and concealed ore targeting, necessitating the application of geophysical exploration to delineate prospective targets. Notably, geophysical surveys have played a defining role in the discovery and subsequent delineation of many nickel, copper, and platinum group element (Ni-Cu-PGE) deposits (Balch et al., 2010).

To effectively support deep and peripheral exploration in the mining area and maximize the indirect ore-targeting capabilities of geophysical methods, this study comprehensively reviews the integrated geophysical exploration approaches implemented at the Xiarihamu. It focuses on characterizing anomaly signatures of diverse geophysical techniques, evaluates their ore-exploration efficacy, and optimizes synergistic method combinations. These efforts not only advance the ongoing exploration of deep-seated and peripheral mineralization but also

establish a referential framework for geophysical exploration strategies targeting analogous deposit types.

Geological Background of Xiarihamu

The Xiarihamu deposit is located at the East Kunlun Orogenic Belt (EKOB), which lies on the northeast margin of the Tibet Plateau (Pan et al., 2017). The EKOB trends east-west and is approximately 1,500 km long and 50 to 200 km wide (Liu et al., 2018). The EKOB was further divided into three regions: the Northern, Central, and Southern East Kunlun Zone, and the Xiarihamu deposit is located at the northern part of the Central East Kunlun Zone (Liu et al., 2024) (Figure 1a). This belt also constitutes a long-lived active metallogenic belt characterized by intense and clustered tectono-magmatic activities (Xu, 2011). Furthermore, the region exhibits intense and recurrent tectonic-magmatic events, providing favorable geological conditions for magmatic Cu-Ni polymetallic mineralization (Du et al., 2017). The Xiarihamu Cu-Ni sulfide deposit, discovered within this belt and genetically associated with mafic-ultramafic intrusions, represents the largest magmatic Ni-Co deposit identified to date in an orogenic setting globally (Zhang et al., 2020b). The discovery of this deposit challenges conventional paradigms regarding the tectonic settings of magmatic sulfide deposits, demonstrating that Cu-Ni mineralization can develop in orogenic belts rather than being restricted to cratonic margins. This finding provides a critical case study for understanding orogenic Cu-Ni metallogeny.

Five mafic-ultramafic intrusions have been identified within the deposit and its periphery (Figure 1c), with No.1 rock mass and No.2 rock mass serving as the principal ore-hosting bodies for the Xiarihamu Ni-Co sulfide deposit (Dacheng et al., 2023). No.1 rock mass, the largest and most mineralized ferrous-ultramafic complex in the system (Figure 1d), measures 1,900 m in length and 700 m in width, covering an area of 1.33 km². This elongate E-W trending body (strike: ~ 60°) is partially concealed by Paleoproterozoic Baishahe Formation strata in its southwestern segment (Zhang et al., 2016a). Petrological analysis reveals a composition dominated by pyroxenite, peridotite, and gabbro, with minor granitic dikes (Zhang et al., 2015). The intrusion hosts 20 identified orebodies, comprising 5 surface-exposed and 15 blind occurrences. Quantified metal resources include 1.18 million tonnes (118×10⁴ t) of

nickel, 238,000 tonnes (23.8×10^4 t) of copper, and 40,300 tonnes (4.03×10^4 t) of cobalt. Over 70% of the metallogenic intrusion contains mineralization (Li et al., 2023b). No.2 rock mass exhibits two surface outcrops with exposed areas of 0.10 km² and 0.15 km², respectively. Its lithology is dominated by gabbro with minor pyroxenite (Zhang et al., 2019b). No.3 rock mass primarily consists of magnesian peridotite, while eclogite and garnet amphibolite have also been identified. No.4 and 5 rock masses are predominantly

composed of magnesian peridotite (Zhang et al., 2016b). Current studies constrain the diagenetic and metallogenic age of the Xiarihamu deposit is the Late Silurian–Early Devonian. The metallogenic environment is interpreted as a post-orogenic large-scale extensional setting during the Caledonian period, with the intrusions forming in an extensional island arc tectonic regime. The magmas originated from relatively high-degree partial melting of the mantle within the East Kunlun magmatic arc (Changyi et al., 2015; Li et al., 2012; Zhang et al., 2015).

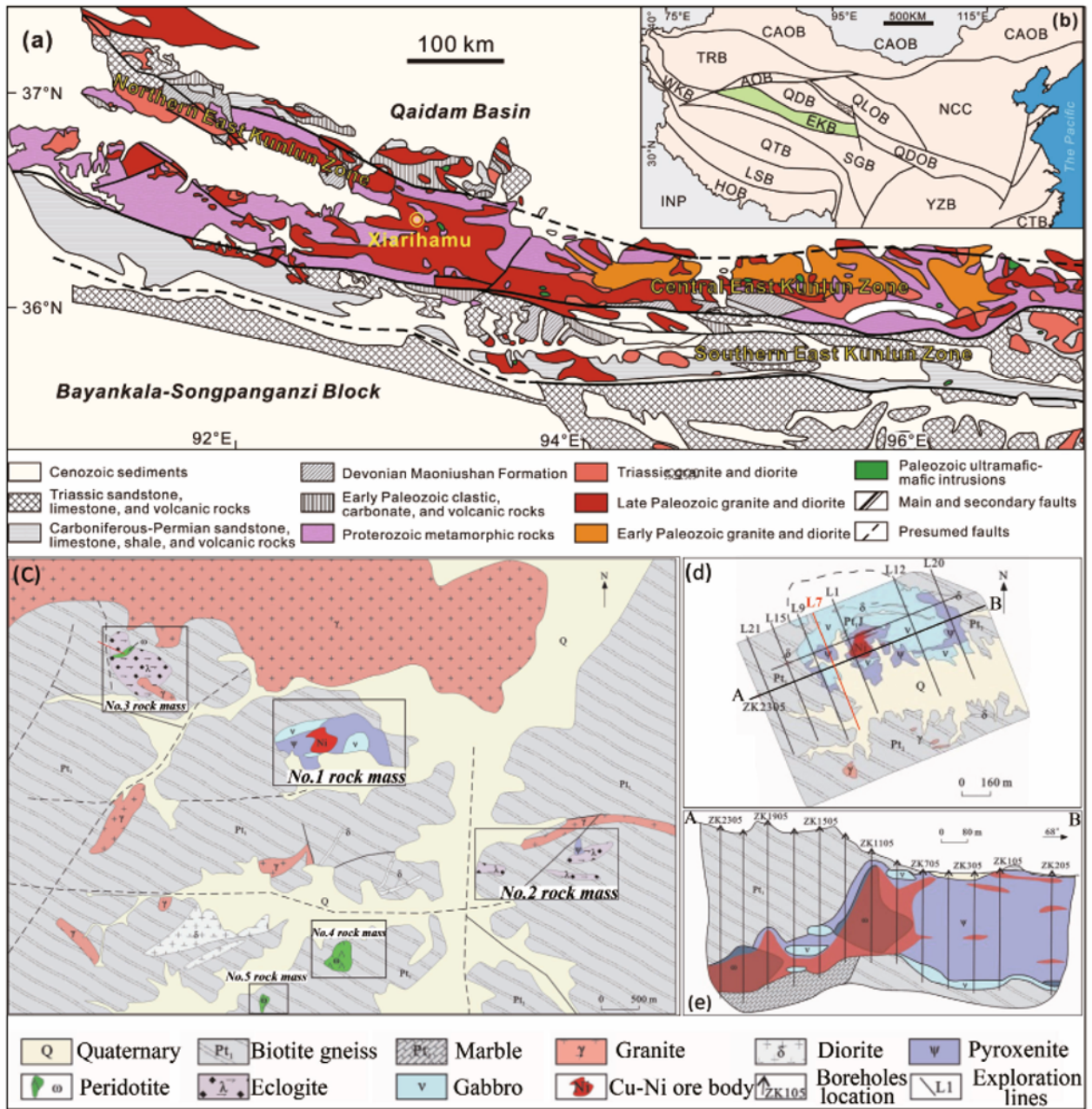


Figure 1. (a) The location and geological settings of East Kunlun Orogenic Belt (EKOB). (b) The geological schematic plot of the EKOB region (Liu et al., 2024). (c) Geological map of Xiarihamu deposit (modified after Li et al. (2023b)). (d and e) No.1 Ore-bearing Rock Mass Plan and Cross-Section of Xiarihamu deposit (modified after Li et al. (2023b)).

Geophysical Background of Xiarihamu

Physical characteristics

The sulphide ores of Ni-Cu-PGE sulphide deposits are strongly anomalous in virtually all physical properties, including electrical conductivity, chargeability, density, magnetic susceptibility, natural radioactivity, and acoustic velocity (King and Milkereit, 2007).

We compiled the physical property data of rock

and ore samples obtained from boreholes along the L7 exploration line in the Xiarihamu mining area, as reported in prior studies. Data analysis revealed that mineralized rocks display characteristic low resistivity, high polarizability, and strong magnetism (Wang et al. 2015), whereas surrounding rocks demonstrate high resistivity, low polarizability, and weak magnetism (Figure 2). These distinct property contrasts provide a robust physical basis for implementing integrated geophysical exploration strategies in the mining district.

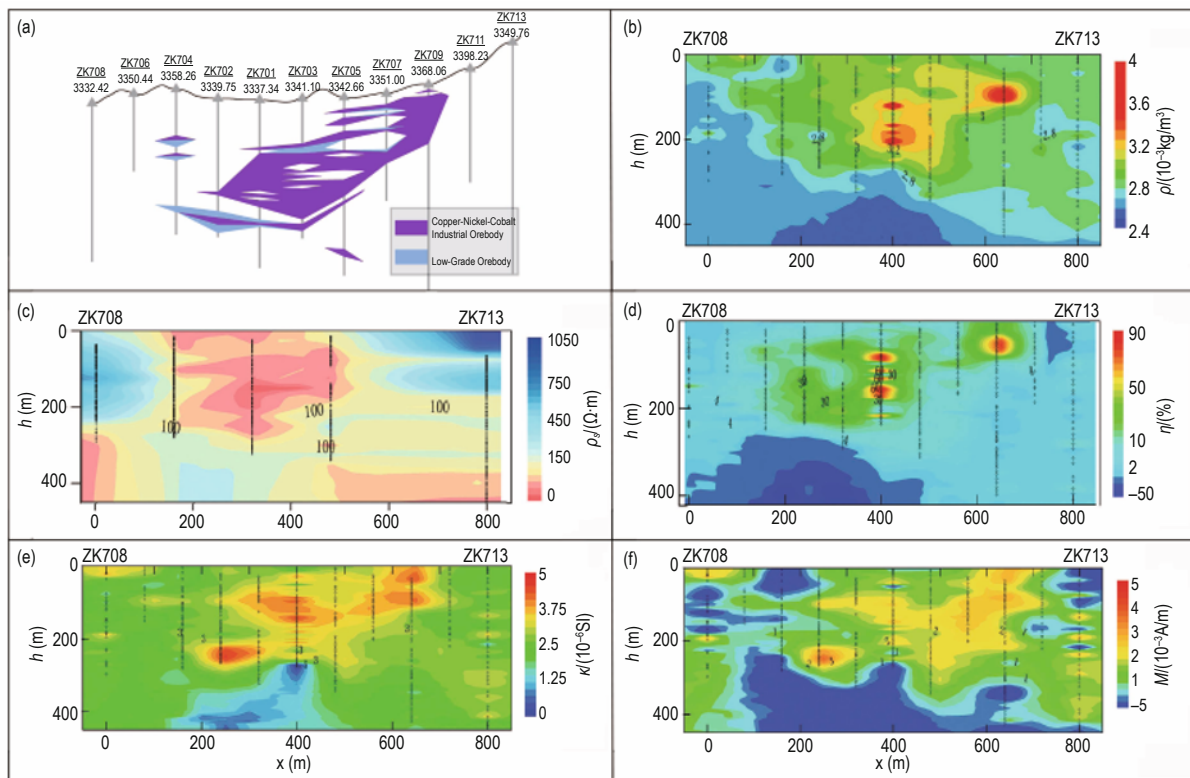


Figure 2. Exploration profile and geophysical property profile of L7 in Xiarihamu (modified after Wang et al. (2015)). (a) The exploration profile of L7. (b)-(f) are the characteristic maps of density, polarizability, apparent resistivity, magnetic susceptibility, and remanent magnetization intensity for the L7 exploration line, respectively.

1:10000 magnetic anomaly

A 1:10,000-scale magnetic survey covers the entire Xiarihamu area. Due to severe topographic constraints hindering the implementation of other geophysical surveys, this 1:10,000 magnetic dataset currently stands as the primary and most extensive geophysical database for the mining district. The 1:10,000-scale high-precision magnetic survey conducted in the Xiarihamu delineated five magnetic anomalies, designated as M-1, M-2, M-3, M-4, and M-5 (Figure 3). Other areas exhibit

stable negative magnetic field zones, with ΔT averaging around - 100 nT and magnetic intensity variations generally below 100 nT. The extensive distribution of metamorphic rocks from the Jinshuikou Group is identified as the primary geological factor contributing to this regional negative magnetic anomaly background (Yan et al., 2016). Positive magnetic anomalies exhibit dispersed distribution patterns, correlating well with the identified mafic complexes within the area (Zhang et al., 2019a). M1 and M5 have relatively concentrated

peak values, indicating typical shallow magnetic bodies (Wang et al., 2024). The segmented appearance of these anomalies is inferred to result from displacement by fault structures through their central sectors. M3, M4, M2, and M6 correspond to No.1 to No.4 mafic-ultramafic complexes discovered in the area, respectively. Among them, No.1 and No.2 complexes commonly exhibit Cu-Ni mineralization, with abundant magnetic minerals such

as pyrrhotite and magnetite present in the ore minerals. The magnetic anomalies are therefore attributed to both the rock masses and magnetic minerals like pyrrhotite and magnetite. Consequently, magnetic surveys have proven to be effective in reflecting mafic-ultramafic complexes and serve as the primary technical method for delineating these rock masses.

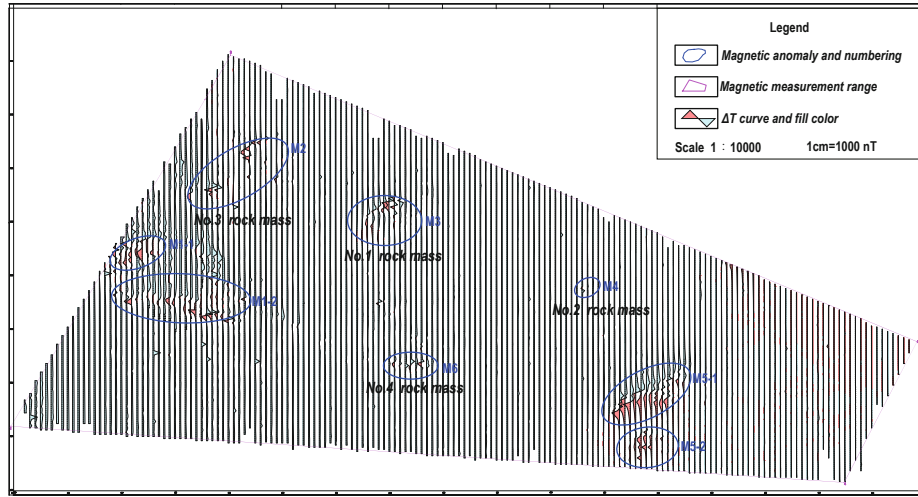


Figure 3. 1:10000 High-precision magnetic ΔT section in the Xiarihamu (modified after Wang et al. (2024)).

Integrated geophysical exploration results

Geophysical explorations have played an important role in the discovery of many magmatic Cu-Ni sulfide deposits (Ma et al., 2024; Xue et al., 2023; Zhang et al., 2020a). Comparing to the surrounding country rocks, sulfide ores possess strong anomaly in many physical properties including electrical property, density and magnetic susceptibility, leading to the geophysical anomalies of higher magnetic force, higher gravity, higher polarizability and lower resistivity, which is the theoretical basis of the traditional geophysical methods for magmatic Cu-Ni sulfide deposits (Yao and Qin, 2014). By studying the physical properties of the principal ore minerals and common associated host rocks, it is possible to understand and hopefully predict the geophysical signatures of most deposits of this type in various geological environments (King and Milkereit, 2007). Therefore, the Xi'an Center of Geological Survey conducted integrated geophysical exploration along exploration line L7 (marked as red exploration line L7

in Figure 1d) of the main ore body in the Xiarihamu No. 1 rock mass, and obtained the geophysical anomaly characteristics of Cu-Ni sulfide deposits in this area.

High-precision gravity measurement

In order to test the effect of gravity measurement at different scales, the point spacing of the test profile was variously sampled at 20 m (1:5000 profile measurement), 40 m (1:10000 profile measurement), and 100 m (1:25000 profile measurement). In recent years, multiple 1:50000 gravity measurements deployed also mostly use this point spacing). Through comparison (Figure 4a-d), it is found that gravity measurement with a point spacing of 100 m can reflect large-scale ore bodies, but the abnormal details are not clear. If it is for a smaller-scale ore body, the gravity anomaly is weakly reflected. Based on the above comparison results, it is recommended that for general mineral exploration, the point spacing of gravity measurement should not be greater than 40 m (1:10000 profile measurement). In a detailed investigation, the point spacing is best selected to be 10 m to 20 m (equivalent to 1:2000 to 1:5000 profile measurement). Otherwise, the abnormally small-scale ore bodies cannot be detected.

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Based on the results of the gravity measurement, the distribution of the Bouguer gravity anomaly in the Line L7 is obtained (Figure 4d). Bouguer gravity anomaly is positively correlated with topography, but there is an obvious high gravity anomaly above the ore body, even though the terrain here is gentle. Therefore, this indicates that it is caused by the ore body with a high gravity anomaly. In order to further reflect the relationship between the gravity anomaly and ore body, we calculated the residual gravity anomaly. First, strata with a depth greater than 2000 m are taken as the background field. Subtracting the background field

from the Bouguer gravity anomaly yields the Bouguer gravity anomaly G1. Secondly, in strata with a depth less than 2000 m, a forward model is established using the measured surrounding rock density and topographic data. The gravity anomaly G2 caused by strata without ore bodies is obtained. Finally, subtracting G2 from G1, the calculated result is the residual gravity anomaly. The residual gravity anomaly is caused by the ore body, and there is an obvious corresponding relationship in spatial position between the two. Under the premise of having density differences, large-scale and high-precision gravity measurement is effective in distinguishing the

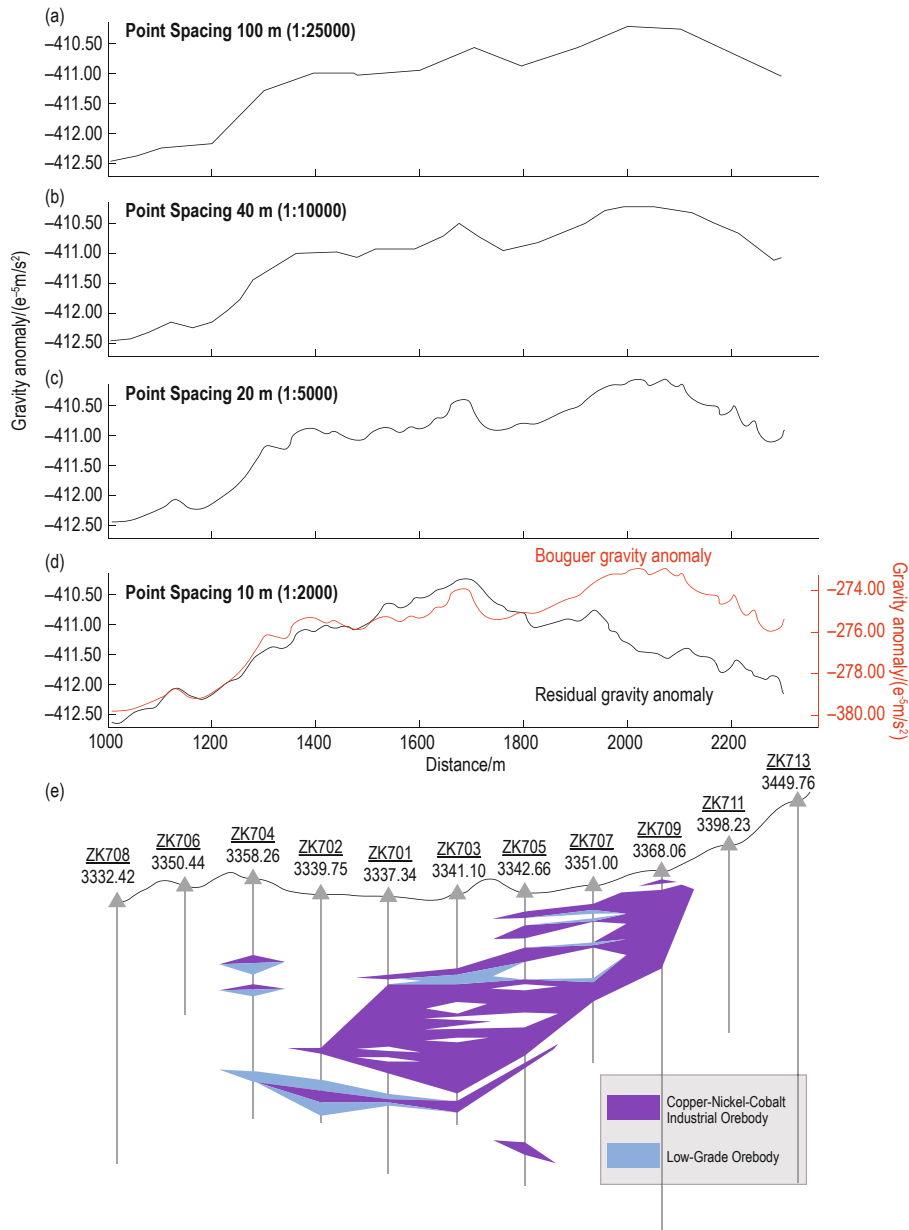


Figure 4. (a)-(c) are the comparisons of gravity measurements at different scales. (d) The comparison of the Bouguer gravity anomaly and the residual gravity anomaly. (e) Geological section of Line L7.

boundary of rock masses. For ore bodies of a certain scale, even the boundary between ore bodies and surrounding rocks can be directly distinguished.

High-precision magnetic measurement

Like gravity measurement, in order to understand the effect of high-precision magnetic measurement at different scales, the interval of test profile points is thinned to 20 m (1:5000 profile measurement), 40 m (1:10000 profile measurement), and 100 m (1:50000 profile measurement or area measurement). Comparing the magnetic anomalies (ΔT) at different scales (Figure

5a-d), for the large-scale ore body of Xiarihamu, the abnormal form of high-precision magnetic measurement with a point spacing of 100 m has completely changed. Therefore, it is completely unreliable to directly deploy drilling verification using 100 m (1:50000 profile measurement or area measurement). It is recommended that for mineral exploration, the point spacing of high-precision measurement should not be greater than 20 m (1:5000 profile measurement). For detailed investigation, the point spacing is preferably 5 m to 10 m (1:1000 to 1:2000 profile measurement).

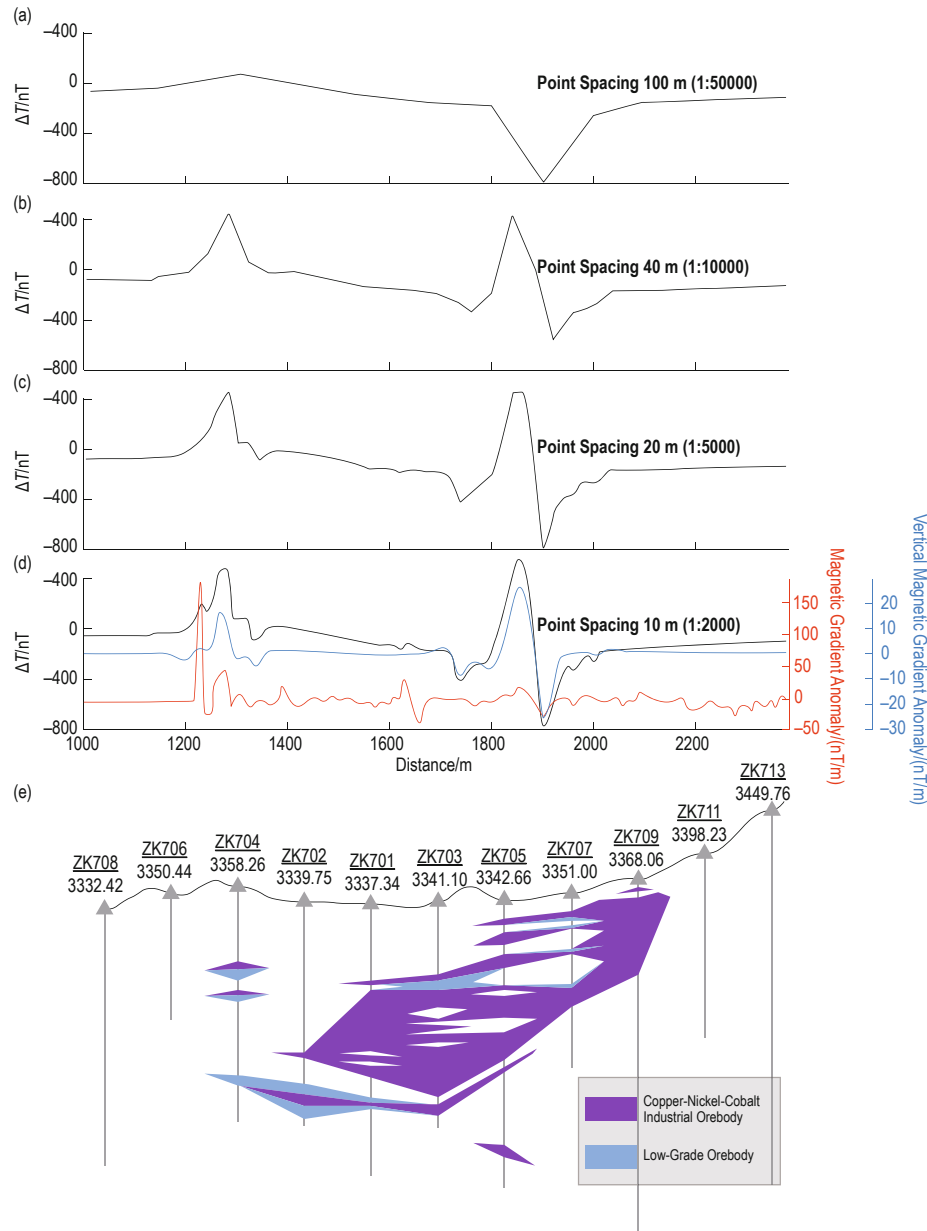


Figure 5. (a)-(c) are the comparisons of magnetic measurements at different scales. (d) The comparison of magnetic, magnetic gradient and vertical magnetic gradient anomalies. (e) Geological section of Line L7.

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The magnetic field differences caused by rocks (ores) are the physical property principles of magnetic exploration. As can be seen from the high-precision magnetic measurement (total misfit of magnetic ≤ 5 nT) profile (Figure 5d), magnetic and the vertical magnetic gradient anomalies are only clearly reflected in the shallow parts where the ore-bearing rock mass is buried. The magnetic anomaly characteristics of the deeper buried parts have little difference from the surrounding rocks. This has a great relationship with the fast attenuation of the magnetic field and the characteristic that magnetic anomalies mainly reflect shallow anomalies. There is no abnormal display in the magnetic gradient measurement. Gradient measurement is not suitable for rock ore bodies with similar medium magnetic anomalies. Combined with previous high-precision magnetic measurement (Ma et al., 2024), high-precision magnetic measurement is effective in finding rock masses at small scales. However, due to factors such as terrain and remanent magnetic direction, the geomagnetic field morphology obtained by ground high-precision magnetic measurements becomes complex. If the magnetic survey results are not processed and analyzed, it is insufficient to deploy drilling only based on the magnetic anomaly.

Induced polarization

Due to the electrical property differences between Cu-Ni sulfide and surrounding rocks, electromagnetic sounding methods are widely used in the exploration of concealed Cu-Ni sulfide deposits. Represented by the Transient Electromagnetic (TEM) method that has been widely applied in recent years, scholars have delineated a series of mineralized rock bodies according to the resistivity profile and carried out drilling verification. It has been proved that through electrical anomaly analysis, favorable exploration targets can be effectively delineated, deep ore-bearing characteristics can be evaluated, and rock body facies belts can be divided. It is also inferred that the low-resistivity anomalies in the electrical profile may also be related to regional faults and deformation zones (Yong-Bang et al., 2016). After a large amount of work has been carried out, the exploration laws and experience of electromagnetic methods for Cu-Ni sulfide ore bodies are continuously summarized and tested. At present, it is generally believed that Cu-Ni sulfide ore bodies are mainly related to medium and low resistivity and high polarizability anomalies.

In the study area, the induced polarization (IP) intermediate gradient measurement is carried out, and tests with different receiving electrode spacings are conducted (Figures 6a and 6b). As the receiving electrode spacing increases, the curve becomes gentler, the abnormal amplitude also decreases, and the details of the anomaly are also not clear. However, if the receiving electrode spacing is too small, the received signal becomes weak and unstable, and false anomalies may occur. Therefore, it is recommended that a receiving electrode spacing test be conducted in the exploration. In the inspection, choose a small receiving electrode distance while ensuring signal stability.

In the study area, tests with different power supply cycles were carried out (Figure 6c). For the abnormal area of polarizability, when the power supply cycle is extended, the abnormal amplitude increases significantly, while in the background area, the polarizability amplitude changes little with the extension of the power supply cycle. As the power supply time is extended, the amplitude increases significantly, while in the background area, the polarizability amplitude changes little with the extension of the power supply cycle. As the power supply time is extended, the charging saturation of sulfide rock (ore) bodies increases, and the polarizability signal becomes more obvious. Therefore, it is recommended to use a long power supply cycle, and the power supply cycle should generally not be less than 16 seconds. Combined with the result analysis of different receiving electrode spacings and different power supply cycles, we propose the following reasons for the ineffectiveness of IP in the study area in the past. Firstly, the transmission power is not enough, and there are problems with the grounding of the transmitting and receiving electrodes, so a sufficiently strong signal cannot be obtained. Secondly, the selection of the receiving electrode spacing is inappropriate. Finally, the power supply cycle is too short.

According to the above experimental results, the power supply electrode distance of IP is 1800 m and the power-off delay is 200 ms (Figure 7a). At the location of the ore-bearing rock mass, the polarization rate (η) increases significantly, and the resistivity decreases significantly. Moreover, the changes in polarization rate and resistivity show a strong correlation with the ore-bearing rock mass. The IP method is a very effective exploration method for polymetallic deposits (especially sulfide deposits) in Xiarihamu. In some areas with weak polarization rate anomalies, prolonging the power

supply period can effectively amplify the abnormal area signal. At the same time, the receiving electrode spacing should not be too large. Under the condition of ensuring

the stability and reliability of the primary field, a small receiving electrode spacing should be selected as much as possible.

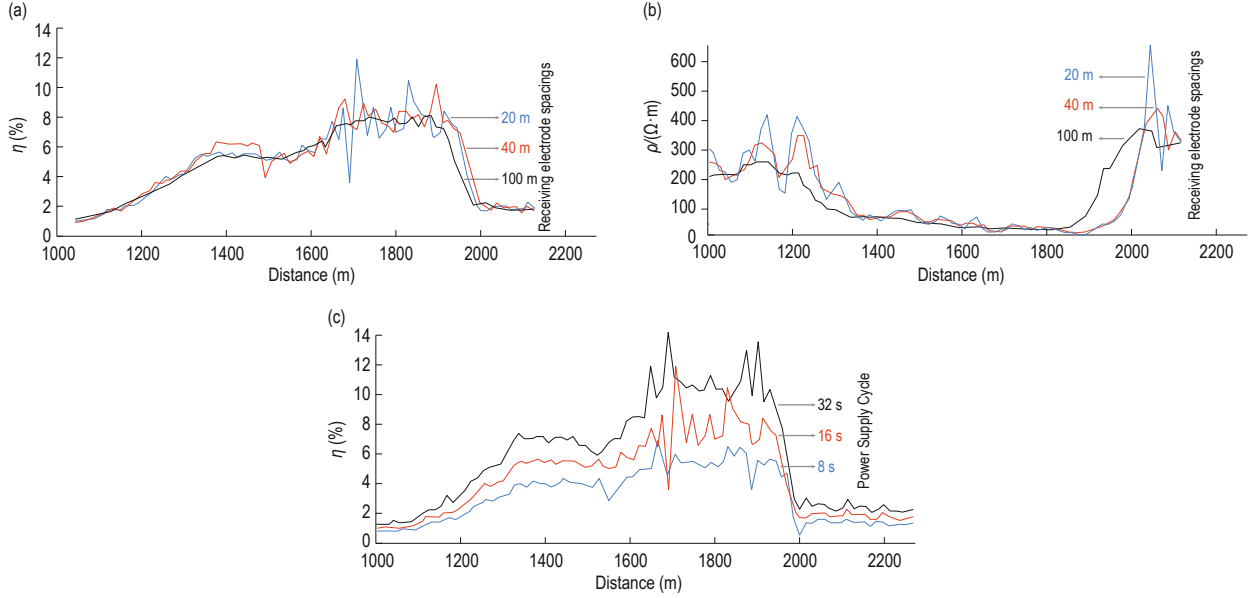


Figure 6. (a)-(c) are the comparisons of induced polarization (IP) measurements at different receiving electrode spacings and power supply cycles.

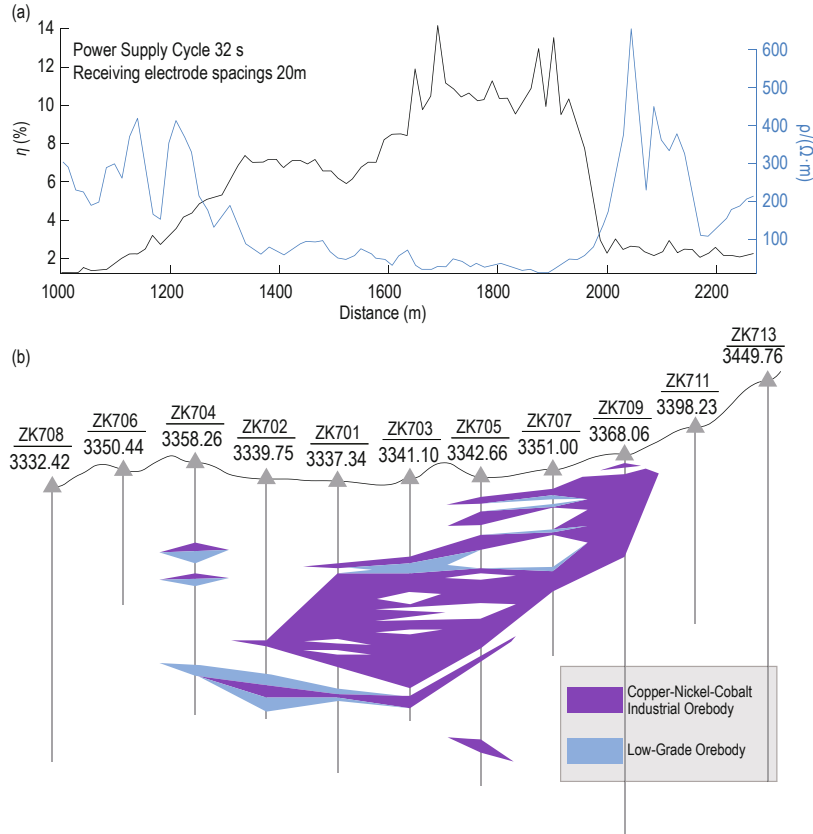


Figure 7. (a) The polarization rate (η) and resistivity (ρ) of induced polarization (IP) measurements. (b) Geological section of Line L7.

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Controlled Source Audio-Frequency Magnetotellurics method

The results of the Controlled Source Audio-Frequency Magnetotellurics (CSAMT) method on the test profile are shown in Figure 8. The resistivity based on the CSAMT method has an obvious corresponding relationship with the ore-bearing rock mass. In addition, under the condition of less interference, the reflection coefficient has obvious interbedded characteristics above the ore-bearing rock mass, and the position where the positive and negative values of the reflection coefficient change is located at the position where the resistivity is 10 Ωm . Abnormal areas of resistivity and reflection coefficient appear at the positions of ZK705 and ZK707, reflecting the interbedded morphology of ore body and non-ore body. It is worth noting that in the inversion

result map of the reflection coefficient, an abnormal area also appears below ZK704, accurately reflecting the existence of a deep blind ore body of ZK704. This contributes to the adjustment of the structural occurrence of ore-bearing rock strata. The CSAMT method has obvious effects on the ore body of Xiarihamu and is an optional exploration method. In the exploration, the following problems should be noted: First, the grounding problem of the transmitting electrode should be handled well to minimize the grounding resistance and increase the power supply current. Secondly, a high-power generator should be selected as much as possible for the power supply to increase the signal strength and suppress interference. Finally, the transmitting distance should be increased as much as possible to eliminate the influence of near-field effects.

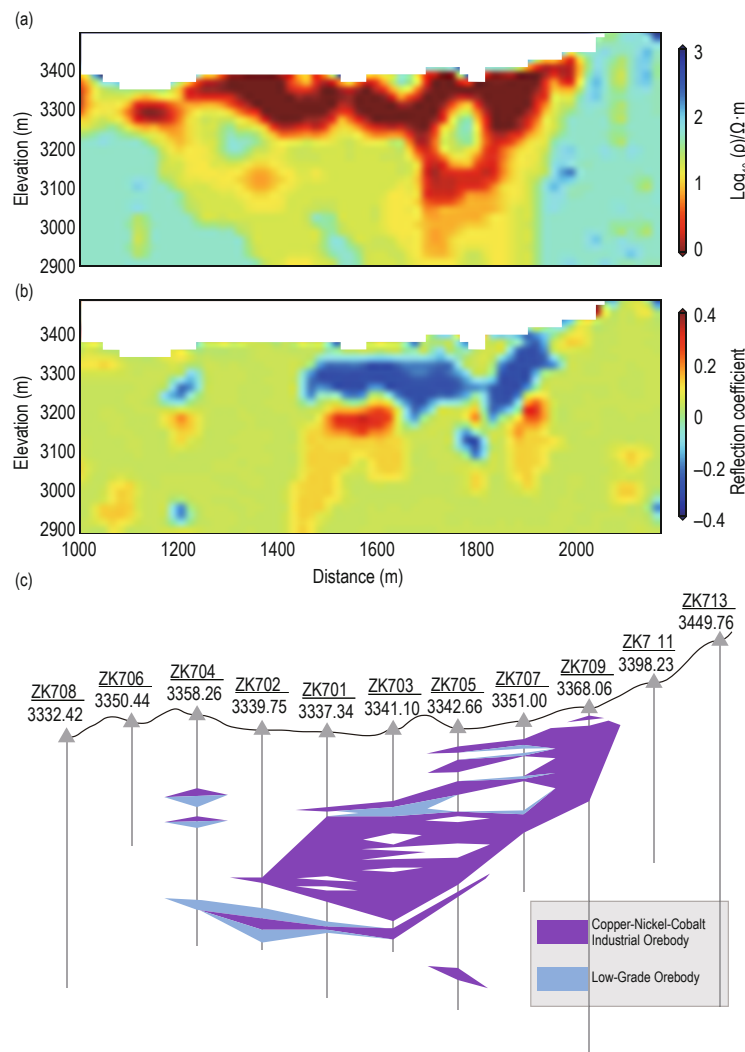


Figure 8. (a) The resistivity profile based on the Controlled Source Audio-Frequency Magnetotellurics (CSAMT) method. (b) The reflection coefficient profile based on the CSAMT method. (c) Geological section of Line L7.

Transient Electromagnetic method

The line L7 of the Transient Electromagnetic (TEM) method is 1000 m. Comparing the resistivity profile with the geological section, the electrical distribution basically matches the geological section well (Figure 9). Near the ZK707 drilling hole, the interbedded form of ore body and non-ore body is reflected on the profile, effectively verifying the effectiveness of the TEM method in this area. The resistivity inversion

profile shows that the buried depth of the low-resistivity anomaly gradually increases from north to south. Moreover, the resistivity of the shallow ore body at the northern end is relatively small. The electrical distribution characteristics correspond well to the distribution form of Cu-Ni mineralized bodies. The low-resistivity anomaly corresponds to high-grade nickel ore bodies. Combined with drilling data, high-grade nickel ore bodies often correspond to low-resistivity anomalies.

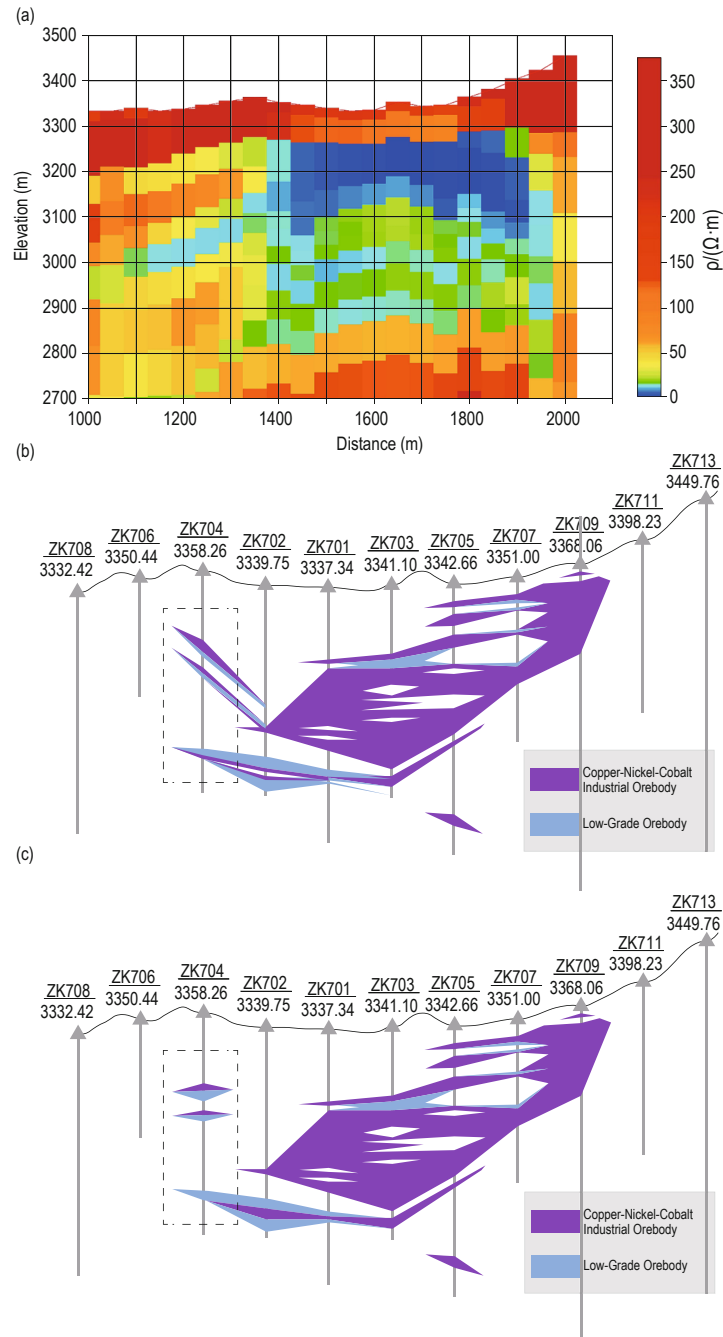


Figure 9. (a) The resistivity profile based on the Transient Electromagnetic (TEM). (b) Geological section of Line L7 before adjustments. (c) Geological section of Line L7 after adjustments.

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The ore-bearing rock formation had a “basin-shaped” distribution based on drilling and trenching results (Li et al., 2021; Wang et al., 2024). However, later TEM inversion profiles showed that the low-resistivity anomaly on the northern side dipped gently and was quite thick, while on the southern side, it inclined southward and gradually tapered off. The resistivity model differs from the geological exploration profile. Geologists then integrated the TEM inversion results with drilling data to make appropriate adjustments to the structural attitude of the deep-seated ore-bearing rock formations (Figure 9b to Figure 9c). It is clear that the TEM inversion results effectively improved the geological understanding of the ore-bearing rock formation's structural attitude. According to the geological and TEM inversion results, the orebody of L7 line has a sediment, while low-resistivity anomaly of the CSAMT inversion is directly exposed to the surface. However, for the geophysical results in the 3250 m deeper area, each method corresponds well.

Conclusions

This paper lists the geophysical characteristics of the Xiarihamu super-large Cu-Ni sulfide deposit. According to the results of physical measurement, high-precision magnetic measurement, IP, high-precision gravity measurement, CSAMT and TEM methods, the mining area shows the geophysical characteristics of “three highs and one low”. That is, the Xiarihamu Cu-Ni sulfide deposit has the characteristics of high gravity, high magnetic, high polarization rate and low resistivity.

Using geophysical methods for exploration has regularity. On the premise of the physical property differences of the Xiarihamu Cu-Ni sulfide deposit, area magnetic measurements can delineate ore-bearing rock masses. However, High-precision magnetic measurements are generally less effective and are only effective for shallow ores. High-precision gravity measurement and the IP method can reflect the boundary between the rock mass and ore body. The electromagnetic method can reflect the shape of the ore body.

Under the conditions of the plateau desert area of the Xiarihamu super-large Cu-Ni sulfide deposit, the following technical method process is proposed. The combination of geophysical methods includes: “geological measurement + soil measurement + magnetic

measurement + high-power IP method + process verification”. First, use geological measurements, 1:10,000 magnetic measurements and soil measurements to narrow down the exploration target area. Secondly, use large-scale magnetic, gravity, and electrical comprehensive profiles and surface mapping methods to locate and delineate the distribution of complex ore-bearing rock masses. Finally, use trenching and drilling to delineate ore bodies. Especially for the concealed rock masses in the covered area, use large-scale, high-precision magnetic and gravity measurements to delineate the ore body range, and the CSAMT and TEM methods to guide the layout of drilling holes, which has achieved obvious results.

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